

# Inventory and Production Management in Supply Chains

Fourth Edition



Edward A. Silver • David F. Pyke • Douglas J. Thomas

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# Preface

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Dramatic advances in information and communication technologies have allowed firms to be more closely connected to a broad and global network of suppliers providing components, finished goods, and services. Customers want their products quickly and reliably, even if they are coming from another continent; and they are often willing to look for other suppliers if products are frequently delivered late. As a result, inventory management and production planning and scheduling have become even more vital to competitive success.

Inventory management and production planning and scheduling have been studied in considerable depth from a theoretical perspective. Yet, the application of these theories is still somewhat limited in practice. A major gap has existed between the theoretical solutions, on the one hand, and the real-world problems, on the other.

Our primary objective in the first three editions of the book was to bridge this gap through the development of *operational* inventory management and production planning decision systems that would allow management to capitalize on readily implementable improvements to current practices. Extensive feedback from both academicians and practitioners has been very gratifying in this regard. Our primary objective is unchanged in this revised edition. In the third edition, we placed an emphasis on how computing tools, such as spreadsheets, can be used to apply the models presented in this book. While sophisticated supply chain planning software is now widely available, we still find that spreadsheets are ubiquitous in practice. As such, we have retained and expanded content on the spreadsheet application of the models in this book.

Advances in information technology have made it easier for firms to exchange information with their trading partners. Indeed, much of the academic and practitioner literature on supply chain management has focused on how firms can collaborate and share information to reduce costs in their shared supply chains. Generally speaking, these initiatives have had a positive economic impact. As discussed in Chapter 1, inventory as a percent of sales, and logistics costs as a percent of gross domestic product, have been consistently declining. Our opinion is that these savings are just the tip of the iceberg. It is now commonplace for an employee in an inventory or manufacturing planning role to (a) have powerful tools on their personal computer, and (b) have access to demand and supply-related data. This means it has never been easier to implement rigorous planning models such as those presented in this book. This is not to say that it is easy, just *easier* than before. There are new challenges to planning in modern supply chains where operations are distributed throughout the globe and across many companies. As such, we have included new material in this edition addressing how to collaborate from a planning perspective across departments within an organization and with external trading partners.

As with all editions of this book, we have drawn on the writings of literally hundreds of scholars who have extended theory and who have implemented theory in practice. We have also incorporated many helpful suggestions from our colleagues.

Where appropriate, new modeling approaches and an updated discussion of the literature have been added throughout the book. Here, we briefly highlight some of the major changes in this edition.

1. The first two chapters from the previous edition have been updated and streamlined into one chapter. This chapter highlights recent macroeconomic trends in inventory and logistics costs, and includes a discussion of how these changes affect operations and corporate strategy.
2. Chapters from the previous edition on Just-in-Time and short-range production scheduling have been updated and consolidated into a single chapter.
3. A new chapter on coordinating inventory management in the supply chain has been added. This chapter addresses collaboration and coordination within a firm through processes such as sales and operations planning, as well as external collaboration through initiatives such as collaborative planning, forecasting and replenishment, and vendor managed inventory. Contractual approaches for coordinating inventory decisions between firms are also covered.
4. The treatment of how to manage inventory for items with low-volume, erratic, or non-stationary demand has been expanded. This includes the addition of derivations and spreadsheet applications for the Poisson distribution.
5. The discussion of supply chain planning software tools including Enterprise Resource Planning (ERP) systems and advanced planning tools has been updated and expanded.
6. The treatment of managing multiple items through the use of composite exchange curves has been expanded. To accompany this material, data sets and associated problems are available as supplemental material. These data sets provide students with the opportunity to apply the concepts in the book to realistic, practical settings.

We have continued to attempt to provide a deep and rigorous treatment of the material without presenting complicated mathematics in the main text. Additional derivations have been added in this edition, but this content generally appears in an appendix. We have chosen this style of presentation deliberately so that sufficient material of interest is made available to the analytically inclined reader, while at the same time providing a meaningful text for a less analytically oriented audience.

Section I of the book presents a discussion on inventory management and production planning decisions as important components of total business strategy. The topics covered include operations strategy, the diverse nature of inventories, the complexity of production/inventory decision making, cost measurement, and an introduction to the important concept of exchange curves. A separate chapter is devoted to the determination of forecasting strategy and the selection of forecasting methods.

Section II is concerned with traditional decision systems for the inventory control of individual items. The use of approximate decision rules, *based on sound logic*, permits realistic treatment of time-varying (e.g., seasonal) demand, probabilistic demand, and the many different attitudes of management toward costing the risk of insufficient capacity in the short run.

Section III deals with special classes of items, including the most important (Class A) and the large group of low-activity items (Class C). Also discussed are procedures for dealing with items that can be maintained in inventory for only relatively short periods of time—for example, style goods and perishable items (the newsvendor problem).

Section IV addresses three types of coordination of groups of items primarily in nonproduction contexts. First, there is the situation of a family of items, at a single stocking location, that share a common supplier, a common mode of transport, or common production equipment. The

second type of coordination is where a single firm is managing inventory across several of their locations in a multiechelon network. For such a setting, replenishment requests from one stocking point become part of the demand at another location. The third context is where items are managed across a network where locations in the network are managed by different firms. This setting requires new approaches to share information and align incentives among trading partners to effectively manage inventory throughout the supply chain.

Section V is concerned with decision making in a production environment. A general framework for such decision making is presented. It covers aggregate production planning, material requirements planning, enterprise resource planning systems, JIT, OPT<sup>®</sup>, and short-range production scheduling.

This book should be of interest to faculty and students in programs of business administration, industrial/systems engineering, and management sciences/operations research. Although *the presentation is geared to a foundational course in production planning, scheduling, and inventory management*, the inclusion of extensive references permits its use in advanced elective courses and as a starting point for research activities. At the same time, the book should continue to have a broad appeal to practicing analysts and managers.



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Many of our professional colleagues have provided helpful comments concerning our research papers, the earlier editions of the book, and the drafts of the current edition. In addition, numerous other colleagues have made available drafts of papers in the more advanced topic areas. Many of the publications of these contributors are referenced in this book. We appreciate all these important contributions. The list of all these colleagues would be too long to include, but we do wish to specifically mention Daniel Costa (Nestle S.A.), Robert Lamarre (Gestion Conseil Robert Lamarre), and David Robb (University of Auckland).

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**Edward A. Silver**  
**David F. Pyke**  
**Douglas J. Thomas**



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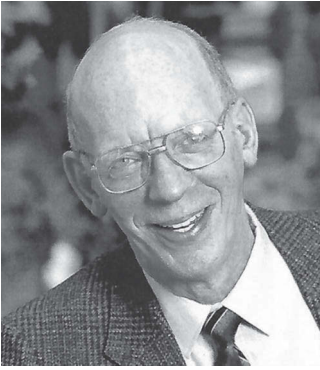
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Professor Silver has presented seminars and talks at national and international meetings of a number of professional societies as well as educational institutions throughout North America and in parts of Europe, Asia, and New Zealand. He has published close to 170 articles in a broad range of professional journals. Dr. Silver has also served in an editorial capacity for several journals.

Professor Silver spent four years as a member of the Operations Research Group of the international consulting firm, Arthur D. Little Inc. Subsequently he has done independent consulting for a wide range of industrial and government organizations throughout North America and elsewhere. These consulting activities have addressed both tactical and strategic problems arising in the management of operations. Specific areas of application have included inventory management, supply chain management, process improvement, production planning, and logistics management. An additional important activity has been his involvement in several executive development programs and other workshops related to the inventory/production field.

Dr. Silver has received extensive professional recognition. This has included being named as a Fellow of five organizations, specifically the Institute of Industrial Engineers (1995), the Manufacturing and Services Operations Management Society (2000), the International Society for Inventory Research (2000), the Institute for Operations Research and the Management Sciences

(2003), and the Production and Operations Management Society (2010). He also was the recipient of two of the highest awards of the Canadian Operational Research Society, namely the Award of Merit (1990) and the Harold Larnder Memorial Prize (2007). Finally, he was awarded a Visiting Erskine Fellowship at the University of Canterbury, Christchurch, New Zealand (1998) and a Visiting Research Fellowship by the Japan Society for the Promotion of Science (2002).



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His research interests include coordinating production and inventory planning across the extended enterprise and connecting decision models to logistics performance measurement. His work has appeared in several academic and practitioner journals in the areas of logistics and operations management, including *Management Science*, *Manufacturing and Service Operations Management*, and *Production and Operations Management*. He serves as a senior editor for *Production and Operations Management*. He is a member of the Institute for Operations Research and Management Sciences, and the Production and Operations Management Society.



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# THE CONTEXT AND IMPORTANCE OF INVENTORY MANAGEMENT AND PRODUCTION PLANNING

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I

Since the mid-1980s, the strategic benefits of inventory management and production planning and scheduling have become obvious. The business press has highlighted the success of Japanese, European, and North American firms that have achieved unparalleled efficiency in manufacturing and distribution. In recent years, many of these firms have “raised the bar” yet again by coordinating with other firms in their supply chains. For instance, instead of responding to unknown and highly variable demand, they share information so that the variability of the demand they observe is significantly lower.

In spite of the media attention devoted to these developments, two mistakes are far too common. First, many managers assume that new levels of efficiency can be attained simply by sharing information and forming “strategic alliances” with their supply chain partners. These managers do not understand that “the devil is in the details,” and that knowing what to do with the data is as important as getting the data in the first place. Developing sound inventory management and production planning and scheduling methods may seem mundane next to strategy formulation, but these methods are a critical element of long-term survival and competitive advantage. Second, many analysts assume that implementing sophisticated inventory and production methods will solve all the problems. These analysts may achieve a high level of efficiency by optimizing given the lead times and demand variability the firm observes. But they do not understand that this efficiency is insignificant compared to the efficiencies available by *changing the givens*.

In Chapter 1, we discuss the importance of inventories and production planning at an aggregate level. This includes a discussion of the role of aggregate inventories in the business cycle. We also present an overview of corporate strategy, and discuss the linkages of finance and marketing strategies with inventory management and production planning and scheduling. We then present a framework for formulating an operations strategy, and we describe how inventory management and production planning and scheduling fit in that context.

Chapter 2 presents several frameworks that will be helpful to operating managers, and to general managers. We describe various statistical properties of inventories, and we discuss how inventories can be classified. A useful framework for understanding different production processes is presented. Finally, we discuss the costs involved, and briefly note where they can be found.

Since managerial expectations about the future have such a tremendous impact on inventory management and production planning and scheduling decisions, forecasting the demand variable is given special treatment in a separate chapter. In Chapter 3, we present a number of strategies and techniques that could be adopted to cope with the unknown future.

## *Chapter 1*

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# **The Importance of Inventory Management and Production Planning and Scheduling**

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Some of the strongest and fastest-growing industries throughout the world are in the service sector. The consulting and financial services industries, in particular, hire thousands of college graduates each year. This fact has prompted many to suggest that manufacturing in developed countries has not only declined in importance, but is on a path toward extinction. It seems clear, however, that nations care deeply about manufacturing. This is particularly true in certain industries. Our observations suggest that, although they receive little attention in the press, the textile/apparel, machine tool,\* and food industries are considered vital. Why is this so? We think it is due to at least two reasons. The first is that food and machine tools are fundamental to national security. Without an industry that manufactures and distributes food, a nation is very vulnerable if it is isolated due to a conflict; and in such situations, without machine tools, the hardware that drives the economy will grind to a halt. The second reason is employment. The textile/apparel industry employs millions of people worldwide, many in low-skill, entry-level, jobs. Without this industry a nation can face high levels of unemployment and has reduced capacity to bring people, such as immigrants, into the workforce.

Many other industries including microprocessors, computers, and automobiles are also considered vital in today's world. These industries are often the source of innovation, productivity improvements, and high-skill jobs. Consider, for example, the productivity improvements that resulted from the introduction of computer numerically controlled machine tools. So, while services have been growing in importance, we submit that manufacturing is still fundamental to the health of most modern economies. Adoption of the methods discussed in this book will ultimately help to strengthen both developed and developing economies, to the extent that they are successfully implemented in manufacturing and logistics firms. Many of the methods in this book also

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\* Machine tools are the machines that make machines, and therefore are the fundamental building blocks for any manufacturing operation.

apply to service industries such as banking, hospitals, hotels, restaurants, schools, and so on, where firms hold inventories of supplies. These stocks must be managed carefully, and in some cases, careful management of these stocks can be critical to the performance of the firm. Two recent studies highlight the importance of inventory management to firm performance. Hendricks and Singhal (2009) show that excess inventory announcements are followed by a negative stock market reaction, and Chen et al. (2007) relate abnormal inventory levels to poor stock returns.

Costs associated with production, inventory, and logistics are quite economically significant. Table 1.1 illustrates logistics costs in the United States. Note that the value of inventories in U.S. firms exceeded \$2 trillion in 2007, and increased to close to \$2.5 trillion by 2014. Total logistics costs, expressed as a percentage of gross domestic product (GDP), have slowly decreased in recent years, from 9.9% in 2000 to 8.2% in 2014. Several key factors continue to make effective inventory management challenging, including increasing product variety, shortening product lifecycles, and an increase in global sourcing. Despite these pressures, inventory levels for manufacturers have been dropping. Figure 1.1 shows inventory-to-sales ratios over time for U.S. retailers, manufacturers, and wholesalers. The figure suggests that inventory levels were decreasing up until 2009, perhaps starting to increase since then. In a rigorous examination of U.S. firms between 1981 and 2000, Chen et al. (2005) report that inventory levels dropped at an average annual rate of 2%. The majority of this improvement came from a reduction in work-in-process (WIP) inventory, with finished goods inventories not declining significantly. In a related study, Rajagopalan and Malhotra (2001) also find that raw material and WIP inventories declined for U.S. firms between 1961 and 1994.

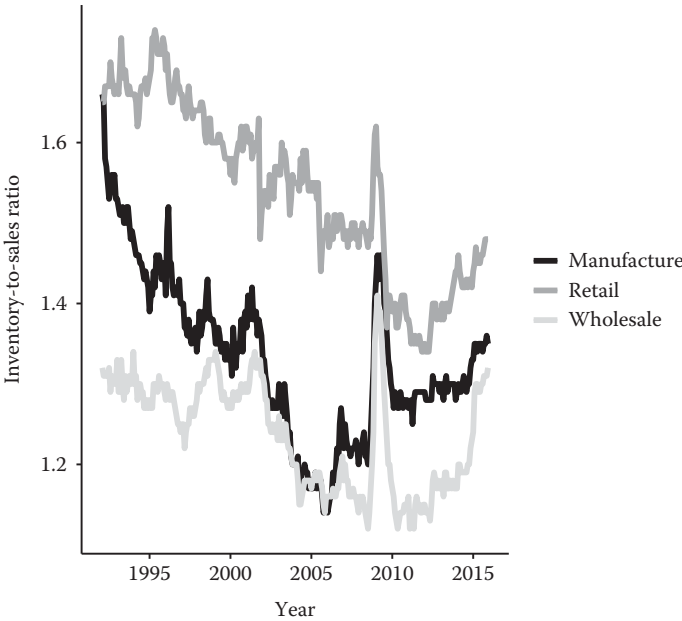
While inventory levels as a percentage of sales have been declining, inventories have shifted downstream, closer to the customer. Figure 1.2 shows the fraction of inventory held by manufacturers compared to the fraction held by wholesalers and retailers. In 1992, manufacturers held 46% of total inventory compared to 37% in 2014. There are several factors that may be causing this shift. In recent years, firms have sourced their products from all over the globe in an effort to find cost efficiencies. Global sourcing initiatives will invariably increase lead times leading to higher inventory levels for wholesalers and retailers. A recent empirical study by Jain et al. (2013) reported that firms with more global sourcing have higher inventory levels. In addition to global sourcing initiatives, product variety offered to customers continues to increase. Firms with many products and short product lifecycles often have high gross margin and high demand uncertainty (Fisher 1997). An empirical study by Rumyantsev and Netessine (2007) confirms that higher gross margin and more demand uncertainty translate to higher inventory levels. An empirical study of U.S. retailers by Kesavan et al. (2016) observes that retailers with high inventory turnover are able to more effectively adjust purchase quantities to react to demand fluctuations, and the negative consequences of excess and shortage of inventory are far less severe for retailers with high inventory turnover.

For analysts, managers, consultants, and entrepreneurs, the opportunities to add value to manufacturing or logistics firms are enormous. On the basis of research and our own experience, it is evident that most firms do not fully understand the complexities of managing production and inventory throughout their supply chain. While there are certainly opportunities for the creative manager in the realms of finance and marketing, our focus is on the benefits that can be won by careful and competent management of the flow of goods throughout the supply chain. One of the authors regularly requires students in an inventory management class to work on consulting projects with local firms. Over the years, we have seen that in more than 90% of the cases, improved inventory or production management would lead to cost savings of at least 20%, without sacrificing customer service. This figure has been replicated in our consulting experience as well. One of our

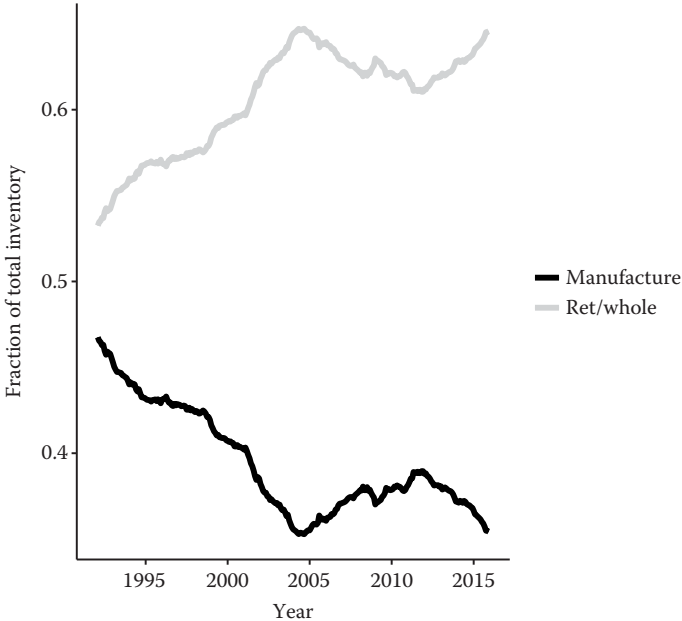
**Table 1.1 U.S. Logistics Costs (\$ Billion)**

| <i>Year</i> | <i>GDP</i> | <i>Value of<br/>Business<br/>Inventory<br/>(a)</i> | <i>Inventory<br/>Carrying<br/>Rate (%)<br/>(b)</i> | <i>Inventory<br/>Carrying<br/>Costs<br/>(c) = (a × b)</i> | <i>Transportation<br/>Costs<br/>(d)</i> | <i>Administrative<br/>Costs<br/>(e)</i> | <i>Total U.S.<br/>Logistics<br/>Costs<br/>(f) = (c + d + e)</i> | <i>Total<br/>Logistics<br/>Costs as a<br/>% of GDP</i> |
|-------------|------------|----------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------|-----------------------------------------|-----------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------|
| 2000        | 10,280     | 1,525                                              | 25.3                                               | 386                                                       | 594                                     | 39                                      | 1,019                                                           | 9.9                                                    |
| 2001        | 10,620     | 1,448                                              | 22.8                                               | 330                                                       | 609                                     | 37                                      | 976                                                             | 9.2                                                    |
| 2002        | 10,980     | 1,495                                              | 20.7                                               | 309                                                       | 582                                     | 35                                      | 926                                                             | 8.4                                                    |
| 2003        | 11,510     | 1,557                                              | 20.1                                               | 313                                                       | 607                                     | 36                                      | 956                                                             | 8.3                                                    |
| 2004        | 12,270     | 1,698                                              | 20.4                                               | 346                                                       | 652                                     | 39                                      | 1,037                                                           | 8.5                                                    |
| 2005        | 13,090     | 1,842                                              | 22.3                                               | 411                                                       | 739                                     | 46                                      | 1,196                                                           | 9.1                                                    |
| 2006        | 13,860     | 1,994                                              | 24.0                                               | 479                                                       | 809                                     | 50                                      | 1,338                                                           | 9.7                                                    |
| 2007        | 14,480     | 2,119                                              | 24.1                                               | 511                                                       | 855                                     | 54                                      | 1,420                                                           | 9.8                                                    |
| 2008        | 14,720     | 2,050                                              | 21.4                                               | 439                                                       | 872                                     | 52                                      | 1,363                                                           | 9.3                                                    |
| 2009        | 14,420     | 1,927                                              | 19.3                                               | 372                                                       | 705                                     | 43                                      | 1,120                                                           | 7.8                                                    |
| 2010        | 14,960     | 2,130                                              | 19.2                                               | 409                                                       | 770                                     | 47                                      | 1,226                                                           | 8.2                                                    |
| 2011        | 15,520     | 2,301                                              | 19.1                                               | 439                                                       | 819                                     | 50                                      | 1,308                                                           | 8.4                                                    |
| 2012        | 16,160     | 2,392                                              | 19.1                                               | 457                                                       | 847                                     | 52                                      | 1,356                                                           | 8.4                                                    |
| 2013        | 16,770     | 2,444                                              | 19.1                                               | 467                                                       | 885                                     | 54                                      | 1,406                                                           | 8.4                                                    |
| 2014        | 17,420     | 2,496                                              | 19.1                                               | 477                                                       | 901                                     | 55                                      | 1,433                                                           | 8.2                                                    |

Source: Council of Supply Chain Management Professionals Annual State of Logistics Report® 2015.



**Figure 1.1** Inventory-to-sales ratios (seasonally adjusted) for U.S. retailers, manufacturers, and wholesalers. (From U.S. Census Bureau. November 2015 Manufacturing and Trade Inventories and Sales report.)



**Figure 1.2** Distribution of inventory between manufacture and combined wholesale and retail (seasonally adjusted). (From U.S. Census Bureau. November 2015 Manufacturing and Trade Inventories and Sales report.)

goals in this book is to provide the reader with knowledge and skill so that he or she can bring real value to employers and, as a result, to the economy as a whole.

In Section 1.1, we discuss how business cycles affect aggregate inventory investments. We turn our attention to the connection between corporate strategy and inventory management and production planning in Section 1.2. In Section 1.3, we extend the strategy discussion to the areas of marketing and finance. Section 1.4 contains a framework for operations strategy. Section 1.5 deals with an important ingredient of decision making—namely, the specification of appropriate measures of effectiveness.

## 1.1 Why Aggregate Inventory Investment Fluctuates: The Business Cycle

It is an unfortunate fact that economies go through cycles—periods of expansion when employment rates are high and the general mood is one of unending prosperity, followed by periods of contraction when unemployment grows and there is a general feeling of malaise about the economy (Mack 1967; Reagan and Sheehan 1985; Blinder and Maccini 1991). As we shall see, inventories play an important role in these cycles. Many economists have tried to compile comprehensive models of the business cycle to explain all patterns of fluctuations that have occurred historically (e.g., Schmitt-Grohé and Uribe 2012). To date, no one has succeeded in building an all-purpose model. Nevertheless, it is apparent that although each cycle is somewhat different, especially with regard to its exact timing and relative magnitude, the cycles continue and there are several common underlying factors. These are illustrated in Figure 1.3.

Prior to the peak labeled in Figure 1.3, the economy is expanding and managers are optimistic about future sales. However, managers can be overzealous in their optimism. In fact, at the peak, because of overoptimistic expectations, too many products are manufactured by the economy and cannot be sold. These surplus goods increase aggregate inventories, and so producers start to decrease production levels. Eventually, the rate of sale exceeds the rate of production of goods.

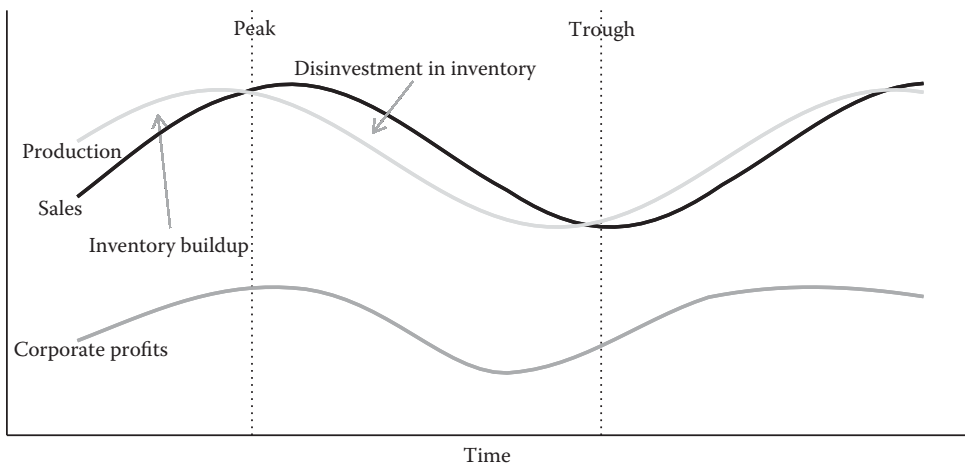


Figure 1.3 The business cycle.

The resulting disinvestment in inventories creates a recession during which prices, production, and profits fall and unemployment is prevalent. The financial crisis of 2007–2008 led to a rapid drop in sales and a subsequent buildup of inventory. This can be seen in the inventory-to-sales ratio of that time period in Figure 1.1.

After a time, an economic recovery is generated by a slowing in the rate of inventory liquidation. Some top managers, expecting that prices will recover, start to slowly expand their operations while costs are low. More and more firms slowly start to hire additional labor, purchase more raw materials, and thereby infuse more money into circulation in the economy. Consumers, with money to spend once again, start to bid up prices of available goods. Once prices of goods start to rise, more and more executives get on the bandwagon by expanding their operations and thereby accentuate the expansionary phase of the cycle. The boom that results eventually is brought to an end when costs of materials are bid up once again by competing firms, when the labor force begins to demand higher wages, and when the scarcity of money for further expansion causes the banks to raise their interest rates. (At this writing, interest rates remain quite low.) The crisis phase that follows is a period of uncertainty and hesitation on the part of consumers and business. Executives find that their warehouses are restocked with excess inventory that, once again, cannot be sold. The business cycle then is ready to repeat itself as explained above.

In Figure 1.3, the following key relationships are illustrated by the data. The peaks and troughs in corporate profits before taxes precede the peaks and troughs in production. Inventory investment lags slightly and thereby contributes to higher cyclical amplitudes in production than are really necessary.

This explanation is, of course, highly simplified, but it does illustrate the main forces, especially the role of inventories, at work during each cycle. Note that the expectations of business and consumers, as well as the ability of decision makers to react quickly and correctly to change, are important determinants of the length and severity of a cycle. Expectations about the future have been shown to depend on the following variables: the trend of recent sales and new orders, the volume of unfilled orders, price pressures, the level of inventories in the recent past, the ratio of sales to inventories (the turnover ratio to be discussed in Section 1.3.1), interest rates on business loans, the current level of employment, and the types of decision-making systems used by management. It is precisely this last point that we address in this book. We want to provide managers with sophisticated, yet understandable approaches for managing production and inventory in the supply chain, enabling them to make good decisions in a timely manner.

Recent research suggests that business cycles are less volatile than they were 50 years ago due, in part, to advances in inventory management and production planning and scheduling (Bloom et al. 2014). Many of the topics we will discuss in this book serve to more closely match production with demand. Therefore, managers can have a more timely perspective on the market, and are less surprised by economic downturns. Other models, such as the probabilistic inventory models discussed in Chapters 6 through 9, have been shown to decrease inventory volatility when applied correctly. Still, there is much work to be done, and there are many opportunities for knowledgeable and creative operations managers to help their firms weather the volatility of economic cycles.

## 1.2 Corporate Strategy and the Role of Top Management

Earlier in this chapter, we saw that, although management of production and inventory throughout the supply chain can be critical to the ability of a firm to achieve and maintain competitive advantage, top managers often do not recognize the importance of these issues. Operations

managers, for their part, often neglect the role their activities play in the strategic direction of the firm, and even in its operations strategy. Both senior and operations managers need to understand the nature of corporate and operations strategy, and they need to understand how inventory management and production planning and supply chain issues impact other functions. In this chapter, we briefly discuss corporate strategy and the functional strategies that derive from it, including marketing, finance, and operations.

It is important to note that some firms have reorganized, or reengineered, to eliminate functional areas and replace them with *business processes*. For instance, a major multinational food and beverage firm recently reengineered in a way that redefined roles to be more responsive to the customer. A common reengineering approach is to replace the operations, logistics, and marketing functions with teams that are process focused. For instance, one team may be devoted to *generating demand*, while another focuses on *fulfilling demand*. The members of the generate demand team perform many of the tasks that traditionally have been done by the marketing department but they may carry out other functions as well, including new product development. The fulfill-demand team often looks like the manufacturing or operations department but may include other functions such as supply chain, logistics, sales, and marketing. The idea is to align the organizational structure with the processes the firm uses to satisfy its customers. Barriers between functions that created delays and tension are removed, enabling the firm to meet customer orders in a more seamless way. For our purposes, the management of inventory, production planning, and supply chains can be thought of as part of the operations function or the fulfill-demand process. In this book, we shall refer to marketing and operations functions rather than to generate and fulfill-demand processes. Nevertheless, the procedures and insights we present apply in either case.

It is not our intention to cover the topic of strategic planning in any depth. There are a number of texts on this topic (e.g., Porter 1980; Mintzberg and Quinn 1991; Grant and Jordan 2015). Instead, our objective is to show that decision making in the production and inventory areas must not be done in a vacuum, as is too often the case in practice, but rather must be coordinated with decisions in other functional areas by means of corporate strategic planning.

The key organizational role of top managers is strategic business planning. Senior management has the responsibility for defining in broad outline what needs to be done, and how and when it should be done. Top management also must act as the final arbiter of conflicts among operating divisions and has the ultimate responsibility for seeing that the general competitive environment is monitored and adapted to effectively. In most business organizations, four levels of strategy can be delineated (listed from the highest to the lowest level):

1. *Enterprise Strategy*: What role does the organization play in the economy and in society? What should be its legal form and how should it maintain its moral legitimacy?
2. *Corporate Strategy*: What set of businesses or markets should the corporation serve? How should resources be deployed among the businesses?
3. *Business Strategy*: How should the organization compete in each particular industry or product/market segment? On the basis of price, service, or what other factor?
4. *Functional Area Strategy*: At this level, the principal focus of strategy is on the maximization of resource productivity and the development of distinctive competencies.

While each level of strategy can be seen as being distinct, they must fit together to form a coherent and consistent whole. Typically, each level is constrained by the next higher one.

We will have more to say on strategic planning in Chapter 13 where a specific framework for decision making in a production environment will be developed.

## 1.3 The Relationship of Finance and Marketing to Inventory Management and Production Planning and Scheduling

### 1.3.1 Finance

Inventories have an important impact on the usual aggregate scorecards of management performance—namely, on the balance sheet and the income statement.\* First, inventories are classified as one of the current assets of an organization. Thus, *all other things being equal*, a reduction in inventories lowers assets relative to liabilities. However, the funds freed by a reduction in inventories normally would be used to acquire other types of assets or to reduce liabilities. Such actions directly influence the so-called *current ratio*, the ratio of current assets to current liabilities, which is the most commonly used measure of liquidity.

Income statements represent the flow of revenues and expenses for a given period (e.g., 1 year). Specifically,

$$\text{Operating profit} = \text{Revenue} - \text{Operating expenses} \quad (1.1)$$

Changes in inventory levels can affect both of the terms on the right side of Equation 1.1. Sales revenue can increase if inventories are allocated among different items in an improved way. Also, because inventory carrying charges represent a significant component of operating expenses, this term can be reduced if aggregate inventory levels decrease. The labor component of operating expenses can also be reduced by more effective production scheduling and inventory control.

A primary aggregate performance measure for inventory management is inventory turnover, or *stockturns*. In this book, we define inventory turnover as

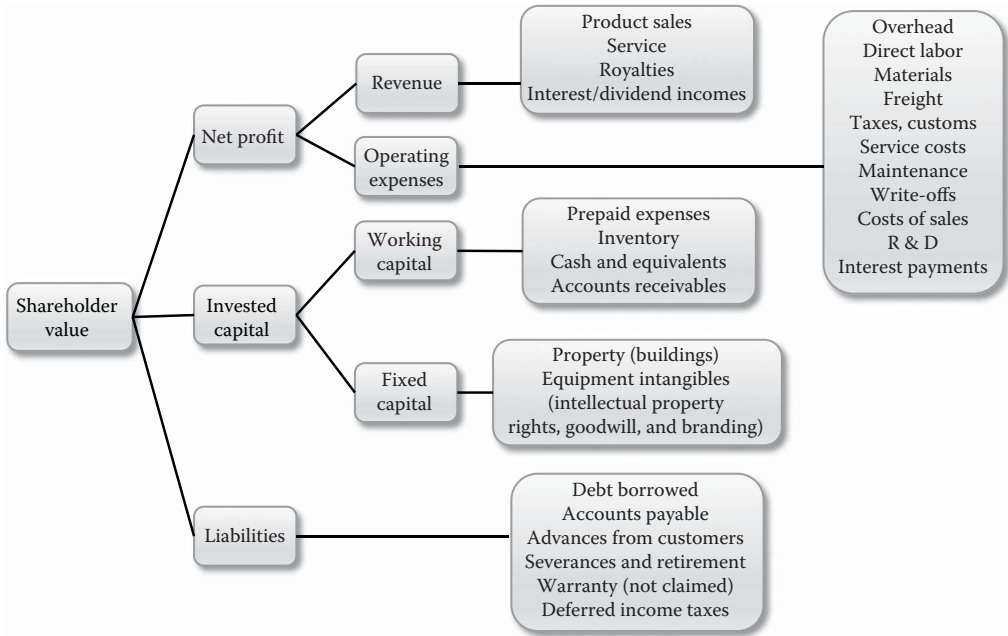
$$\text{Inventory turnover} = \frac{\text{Annual sales or usage (at cost)}}{\text{Average inventory (in \$)}} \quad (1.2)$$

Turnover can be a very useful measure, especially when comparing divisions of a firm or firms in an industry. See Gaur et al. (2005) for a detailed analysis of inventory turnover in the retail sector. From Equation 1.2, we can see that an increase in sales without a corresponding increase in inventory will increase the inventory turnover, as will a decrease in inventory without a decline in sales.

There are several dangers with turnover as a performance measure, however. Too frequently we have seen leaders of manufacturing firms respond to turnover less than the industry average by issuing an edict to increase turnover to the industry average. In one company, while competitive firms purchased some component parts, this company manufactured them because of a special competence in this type of manufacturing. WIP inventory was necessary to keep the production process flowing smoothly. To increase inventory turnover, all they would have to do is to buy more and make less. When the edict came down, the controller recalculated the decision rules for inventory ordering so that lot sizes were cut and raw material inventory would be reduced. Shipments to customers continued for a few months as warehouse inventories were depleted, but then customer service degenerated when finished goods and component inventories bottomed out. Rampant stockouts created massive, expensive expediting. The smaller lot sizes drove up the product cost, and profits plummeted. In short, strategic competitive advantage can have a significant effect on the determination of the appropriate inventory turnover number.

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\* Droms (1979) provides a nontechnical description of financial reports.



**Figure 1.4** Components of shareholder value.

One of the most common measures of managerial performance is the *return on investment* (ROI), which represents the profit (after taxes) divided by the average investment (or level of assets). From the above discussion, it is clear that inventories have an important impact on the ROI (also see Problem 1.18).

Finally we note that many of these relationships can be summarized in Figure 1.4, which is due to Lee, H. and S. Whang (1997, Personal communication). Inventory is a component of working capital, and the interest payments to finance that inventory are a component of operating expenses. Improvements in inventory management, therefore, can have a significant impact on shareholder value. Likewise, improvements in production planning and scheduling can increase customer service and, therefore, product sales, and can decrease expenses related to direct labor, materials, and freight. Thus, net profit can increase, again increasing shareholder value. It is useful to consider other potential impacts of supply chain management, inventory management, and production planning and scheduling on the components of Figure 1.4 (see Problem 1.14).

### 1.3.2 Marketing

The most common complaint we hear from operations managers about marketing is that marketing managers simply do not understand how difficult it is to manufacture and distribute a wide variety of products. Because of low-volume production of many products, productive capacity is lost to setups, and product demand is more difficult to forecast. Workers and equipment must be more flexible, and inventories must be higher. Marketing managers prefer a wide variety of products because they are listening to customers and trying to respond to their needs and desires. High inventories are desirable because demand can be met without delay. Furthermore, sales and

marketing managers may make pricing decisions, such as steep promotional discounts, that increase unit sales, but send a shock through the factory and supply chain. These fundamental conflicts can be mitigated by a process orientation as discussed in the introduction to this chapter. Even with a process orientation, however, conflicts are bound to arise. This again highlights the need for senior managers to be involved in setting policy for supply chain management, inventory management, and production planning and scheduling.

### 1.4 Operations Strategy

In this section, we develop a framework for operations strategy and we emphasize the place of the management of inventory, production planning, and supply chain in the broader context of the operations function.

We begin by stressing again that the operations strategy of the firm should be in concert with other functional area strategies, including marketing, finance, and human resources. Sometimes, however, one function should take precedence over the others. One firm went through a painful period of not responding to customer needs in a long-term effort to improve customer satisfaction. The operations group needed to develop the capability to manufacture high-quality items in high volume. To avoid disruptions during the improvement process, customer desires were neglected in the short term, and marketing was frustrated. In the long term, however, customers were delighted by the level of quality and by the responsiveness of the firm.

The operations strategy itself is made up of three levels: mission, objectives, and management levers. (See Figure 1.5.)

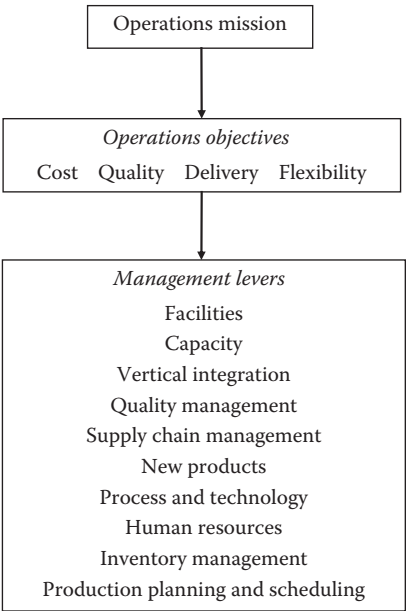


Figure 1.5 A framework for operations strategy.

### 1.4.1 Mission

The operations mission defines a direction for the operations function. McDonald's, for instance, uses four terms to describe its operational mission: quality, cleanliness, service, and value. For many years, the annual report has prominently highlighted these terms, most recently in the 2014 shareholder letter from the CEO. Because the mission should not change significantly over time, the mission statement is often somewhat vague. Otherwise, it is likely that it would have to be reworded frequently as changes occur in the environment, competition, or product and process technology. Employees need to know that there is a consistent direction in which their company is moving.

It is important however that the mission be attractive to excellent people. If it is a sleepy statement of direction that resembles the mission of most other firms in the industry, it will be difficult to attract the best employees, not to mention investors. Therefore, a mission statement should incorporate some of the excitement of top management and should communicate to employees, investors, and customers that this is an excellent firm.

### 1.4.2 Objectives

Because the mission statement is vague, it is difficult to know whether or not it has been achieved. How can one know, for instance, whether "quality" has been achieved? The second level of an operations strategy, operations objectives, provides carefully defined, measurable goals that help the firm achieve its mission. For many years, firms have used four operational objectives: cost, quality, delivery, and flexibility.

It is critical that the objectives are *defined* carefully, clearly *measurable*, and *ranked*. They must be defined carefully because these terms are often used loosely. Quality at a McDonald's restaurant is very different from quality at a five-star restaurant, which is very different from quality in a hospital operating room. It is also important that the objectives be measurable so that management knows whether they are meeting their goals or not. It is quite possible, and perhaps desirable, to use more than one measure for each objective (for instance, warranty cost and parts-per-million defective for the quality objective). The objectives should be rank ordered as well, so that managers can give priority to the correct objective when it is necessary to make tradeoffs. A manager of a high volume manufacturing line made it quite clear to one of the authors that, in his opinion, cost was more important than delivery. When questioned further, however, he noted that in fact he had occasionally gone over budget by using overtime in order to meet a due date. In other words, he said he ranked cost above delivery, but his behavior indicated that delivery was more important. Subsequent discussions with senior managers helped him understand that delivery was, in fact, primary.

In the 1970s, many people in operations thought that cost and quality were opposites, as were delivery and flexibility. For instance, one could aim for low-cost production, but then it would be impossible to achieve high quality. Likewise, if one had rapid delivery, it would be impossible to be flexible. More recent experience, however, suggests that cost and quality are complements rather than opposites. Warranty, prevention, and detection costs decrease as quality improves. Rework and congestion on the factory floor also decrease significantly, thereby reducing the cost of products. In addition, recent experience would indicate that rapid delivery of customized products can be possible. This is especially true with technology such as flexible automation, electronic data interchange, supplier web portals, radio frequency identification (RFID), and scanning technology. Most firms have instances, however, in which tradeoffs among the objectives must be made. Therefore, although combined improvements are possible, the objectives should still be ranked.

The cost objective can be considered in one of the three categories: low, competitive, or relatively unimportant. In a *low-cost* environment, such as that of certain discount retailers, the goal is to have the lowest cost products or services in the industry. Firms that aim for *competitive* costs do not necessarily strive to have the lowest cost products, but rather want to be competitive with most other firms in the industry. Some firms produce prototypes or have a unique product for which they can charge a *premium*; hence, cost becomes less important. Cost measures include dollars per unit, inventory turns, and labor hours per unit (or other productivity measures). In the United States, low cost tended to be the primary objective of manufacturing firms from the 1950s to the mid-1970s.

The quality objective rose to the fore in the mid-1970s to the mid-1980s with the advent of Japanese quality improvements and the sale of Japanese products in the United States. In particular, the automotive industry experienced the stunning effects of high-quality Japanese products. Quality can be defined by understanding which of its multiple dimensions are important. Garvin (1987) describes eight dimensions of quality: performance, conformance, reliability, durability, serviceability, features, aesthetics, and perceived quality. Quality measures include percent defective, percent returns, results from satisfaction surveys, warranty dollars, and so on.

Delivery can be defined on two dimensions, speed and reliability. For instance, some firms compete on delivering within 24 hours of the customer request. Others may take longer but will assure the customer that the products will be provided reliably within the quoted delivery time. Some companies rank delivery last among the objectives. Prototype manufacturing, for instance, sometimes involves complete customization and, therefore, may require long delivery times. Measures for delivery include percent on time, time from request to receipt, fill rate (the percent of demand met from the shelf), and so on.

The fourth objective is flexibility. Flexibility has three dimensions—volume, new product, and product mix. *Volume* flexibility is the ability to adjust for seasonal variations and fluctuations. *New product* flexibility is the speed with which new products are brought from concept to market. Automotive firms in recent years have made great strides in new product flexibility. A niche car allows a firm to jump in quickly to low-volume, profitable, markets. This flexibility is impossible if development time is eight to 10 years, as was traditionally the case with U.S. and European manufacturers. Most manufacturers have now reduced development time to roughly three-and-a-half years, a target first reached by several Japanese automotive firms. *Product mix* flexibility is the ability of a company to offer a wide range of products. This may simply mean that the catalog contains many items, or it may mean that the firm has the ability to develop customized products. Many machine tool companies produce a single product for a given customer, and then never produce that exact product again. Hence, they would rank product mix flexibility very high.

It has been argued that delivery and flexibility are the most important objectives in some industries because cost-cutting programs and quality improvement programs have “leveled the playing field” on cost and quality. These firms then compete on “time”—rapid introduction of new products and rapid delivery of existing products. The phrase *time-based competition* has been used to describe this phenomenon.

Finally, we note that the objectives are dynamic. For instance, as a new product begins full-scale production, the firm may emphasize flexibility (to design changes) and on-time delivery so that market share is not lost to competitors. Over time, as the product design stabilizes, the emphasis may change toward quality and cost.

### 1.4.3 Management Levers

Although the operations objectives provide measurable goals, they do not indicate how a firm should pursue those goals. The 10 management levers—facilities, capacity, vertical integration, quality management, supply chain management, new products, process and technology, human resources, inventory management, and production planning and scheduling—provide the tactical steps necessary to achieve the goals.\*

It is interesting to note that in the late 1970s, researchers did not include quality management, supply chain relationships, new products, and human resources in the list of management levers (Wheelwright 1978). Today, these areas are considered critical to competitive success. It is also certain that new levers will be introduced in the next decade. The categories for the operations objectives, on the other hand, have not changed.

*Facilities* decisions concern the location and focus of factories and other facilities. Where does a company locate its factories and distribution centers to meet the needs of its customers? Is there a need for multiple facilities or is a single location sufficient? Does each facility perform all functions or is one facility focused on a particular market, a particular process, or a particular product? (See Sheu and Krajewski 1996). Many manufacturing companies have one plant that acts as a “parent plant” that is responsible for odd-ball parts and new product introduction. When products reach high-volume manufacture, they are moved to a satellite plant where efforts are focused on excellence in that particular product and/or process. It is critical to evaluate each plant manager based on the specific charter for his or her facility.

*Capacity* decisions involve magnitude and timing of capacity expansion. Capacity decisions interact with the facility location decisions. For instance, some firms set limits on the number of employees that may work at any one location; by remaining smaller than a given size, better communication and teamwork are possible. As demand grows, additional expansion at the same site would violate the limit, and a new plant site must be found. A Massachusetts textile manufacturer had rapidly expanding sales in Europe. When capacity in the Massachusetts factory was no longer able to meet total demand, they had to determine whether to expand in Massachusetts, elsewhere in the United States, or in Europe. The final decision was to build in eastern Germany to exploit lower transportation costs and import duties for European sales, and to take advantage of tax breaks offered by the German government.

*Vertical integration* concerns make/buy decisions. Some firms make components, perform final assembly, and distribute their products. Others focus only on product design and final assembly, relying on other firms to manufacture components and to distribute finished goods. Facilities, capacity, and vertical integration decisions are primarily made for the long term because they involve bricks and mortar and significant investment.

*Quality management* encompasses the tools, programs, and techniques used to achieve the quality goals. It is distinct from the quality objective in that the objective specifies the definition and measurable targets, and the lever specifies the means to achieve the targets. Quality management includes things such as statistical process control (SPC), Taguchi methods, quality circles, and so on. Many aspects of process improvement fall under this management lever, although process improvement leads to increased productivity as well as improved quality. It is important to note that different definitions of quality may dictate the use of different quality procedures.

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\* Portions of this discussion are drawn from Fine and Hax (1985).

The *supply chain management* lever focuses on relationships with suppliers and customers. Indeed, it is concerned with the entire network of firms and activities from, say, digging ore out of the ground, to getting the product into the hands of the customer. Furthermore, it is concerned with the handling of the product after it has finished its useful life. In general, supply chain management uses tools to efficiently move information, products, and funds across the entire supply chain. Chapter 12 contains an extensive discussion of supply chain management.

The *new products* category involves the procedures and organizational structures that facilitate new product introduction. The days of a focused team of engineers working without input from marketing or manufacturing are gone. Most firms now employ multifunctional teams, composed of design engineers, marketing personnel, manufacturing managers, and production line workers. The new product lever specifies the reporting relationships as well as any procedures for setting milestones in the development process.

The *process and technology* category encompasses the choice of a production process and the level of automation. The product-process matrix, to be discussed in Chapter 2, is a useful framework for analyzing process choices. Decisions about implementing new process technology are included here as well.

*Human resources* involves the selection, promotion, placement, reward, and motivation of the people that make the business work.

*Inventory management* encompasses decisions regarding purchasing, distribution, and logistics, and specifically addresses when and how much to order. This management lever is the topic of Chapters 4 through 11, although Chapter 6 alone provides a comprehensive discussion of the issues involved.

Finally, *production planning and scheduling* focuses on systems for controlling and planning production. Chapters 13 through 16 deal with specific production planning and scheduling decisions and systems.

#### **1.4.4 General Comments**

We conclude this section with several general comments. First, researchers and practitioners are continually introducing new approaches addressing some important aspect of management. Inventory management and production planning and scheduling methods that would have been considered impossible just a few years ago are commonplace today. The revolution in information technology and computer networking allows members of a supply chain to have access to manufacturers and to track consumer demand on a daily basis, and thereby more closely match production with demand. Many of these new approaches fall under the board area now widely known as *supply chain management*. The supply chain lever, of course, pertains to supply chain management, but so do inventory management (as we shall see in Chapters 11 and 12), production planning and scheduling, vertical integration, and new products. Occasionally it is necessary to introduce a new lever to focus attention on the important issues, but other times, the existing levers adequately serve the purpose. Even with new supply chain initiatives, however, inventory management and production planning and scheduling continue to be a headache for many managers. Evidence suggests that even though a massive amount of real-time data is available, many firms do not use it, either because they are overwhelmed with data, or because they do not know what to do with it. Although real-time information can significantly reduce the demand variability a firm observes, significant variability and the associated problems usually remain. Managers must understand the value of the information, and they must choose inventory policies and production planning and scheduling systems that are appropriate for their situations. One of our primary goals in this book

is to communicate the basic principles and procedures in this field so that managers, consultants, and analysts can provide insight and value to their firms in this dynamic world.

Second, as a firm audits its manufacturing strategy, it is important to recognize that the policies in place for each of the 10 levers should be consistent not only with the operations objectives, but also among themselves. If there are inconsistencies, the firm should initiate action programs to mitigate any negative effects. For instance, many companies historically pursued quality improvement programs without changing worker incentives. Workers were rewarded for the volume of their output, without regard for quality. In other words, the quality objectives and policies were inconsistent with the human resources policies. Reward systems had to be rearranged to fit with the quality goals. Likewise, as we discuss inventory management and production planning and scheduling in great depth throughout the remainder of this book, keep in mind the relationship of these management levers to the other eight, and to the operations objectives. Extensive effort to reduce inventory cost may not be warranted, for instance, if the firm can charge a premium price and really needs to focus on flexibility. In a similar vein, customer service levels, which will be discussed in Chapter 6, should be established with clear knowledge of the cost and delivery objectives. These issues will be developed further in Chapter 13.

Finally, in the process of auditing a manufacturing strategy, managers should understand distinctive competencies at the detailed level of the management levers. These distinctive competencies should influence the objectives, mission, and business strategy. Thus, information flows both ways, from business strategy to management levers and from levers to business strategy.

## **1.5 Measures of Effectiveness for Inventory Management and Production Planning and Scheduling Decisions**

Analysts in the production/inventory area have tended to concentrate on a single quantifiable measure of effectiveness, such as cost, sometimes recognizing certain constraints, such as limited space, desired customer service, and so forth. In actual fact, of course, the impact of decisions is not restricted to a single measure of effectiveness. The appropriate measures of effectiveness should relate back to the underlying objectives of management. Unfortunately, some of the following objectives are very difficult to quantify:

1. Minimizing political conflicts (in terms of the competing interests) within the organization.
2. Maintaining a high level of flexibility to cope with an uncertain future.
3. Maximizing the chance of survival of the firm or the individual manager's position within the organization.
4. Keeping at an acceptable level the amount of human effort expended in the planning and operation of a decision system.

Moreover, as will be discussed in Chapter 2, even some of the more tangible cost factors are very difficult to estimate in practice.

From the perspective of top management, the measures of effectiveness must be aggregate in nature. Management is concerned with *overall* inventory levels and customer service levels (by broad classes of items) as opposed to, for example, minimization of costs on an individual item basis. Furthermore, rather than optimization in the absolute sense, a more appealing justification for a new or modified decision system is that it simply provides significant improvements (in terms of one or more measures of effectiveness) over an existing system.

One other complexity is that certain decisions in inventory management and production planning and scheduling should not necessarily be made in isolation from decisions in other areas. As an illustration, both Schwarz (1981) and Wagner (1974) point out that there are important interactions between the design of a physical distribution system (e.g., the number, nature, and location of warehouses) and the selection of an appropriate inventory management system. (Chapter 11 will deal further with this particular issue.) Other researchers address another type of interaction, namely, between transportation (shipping) decisions and those of inventory management. (See Ernst and Pyke 1993; Henig et al. 1997.) Another interaction, vividly portrayed by the success of Japanese management systems, is between inventory management and quality assurance. Finally, interactions between pricing and inventory decisions have generated much interest in research and in practice. (See Chan et al. 2004; Fleischmann et al. 2004; Chen and Chen 2015). In this book, we do not develop mathematical models that directly deal with these large-scale interactive problems. However, in several places, further descriptive advice and references will be presented.

## 1.6 Summary

In this chapter, we have examined a number of strategic issues related to inventory management and production planning and scheduling. In much of the remainder of the book, our attention will be focused on specific issues of how best to design and operate an inventory management and production planning and scheduling decision system for a specific organizational context. Nevertheless, it is important not to lose sight of the broader perspective of this chapter.

## Problems

- 1.1 Select two appropriate, recent articles from a newspaper and identify how their contents could impinge on decision making in inventory management and production planning.
- 1.2 Why do inflation and supply shortages tend to increase real inventories? What is the effect of a reduction in the rate of inflation on inventory investment?
- 1.3 Investigate and discuss the inventory control procedures at a small (owner-managed) company. When does a company become too small to have its inventories managed scientifically?
- 1.4 Repeat Problem 1.3 for production planning and scheduling procedures.
- 1.5 Repeat Problem 1.3 for a small restaurant.
- 1.6 Look for evidence in the newspaper or business press to support or contradict the research on business cycles cited in Section 1.1. Discuss your findings.
- 1.7 Find an article in the business press that indicates the position of top management toward manufacturing issues, such as inventory management or production planning and scheduling. Discuss your findings.
- 1.8 Consider any other course you have taken. Briefly discuss how some key concepts from it are likely to interact with decisions in the inventory management or production planning and scheduling areas.
- 1.9 For each of three or four specific local industries, what do you think is the approximate breakdown between raw materials, WIP, and finished goods inventories? Discuss your answer.

- 1.10 In Section 1.5, mention was made of an interaction between the design of a physical distribution system and the choice of an inventory management system. Briefly comment on this interaction.
- 1.11 Examine the interaction specified in Section 1.5 again, but now between the choice of a transportation mode and the selection of inventory decision rules. Briefly discuss your results.
- 1.12 Briefly comment on how inventory management might interact with each of the following functional areas of the firm:
  - a. Maintenance
  - b. Quality control
  - c. Distribution (shipping)
- 1.13 A management decision system designed for providing information on how well a business is doing, relative to other businesses in the same industry segment, is appropriate for
  - a. Top management
  - b. Middle management
  - c. Marketing management
  - d. Factory management
  - e. Office staff

Explain the reasoning behind your choice.
- 1.14
  - a. What components of Figure 1.4 would change, and in what direction, if production planning and scheduling systems were improved?
  - b. What components of Figure 1.4 would change, and in what direction, if inventory management systems were improved?
  - c. What components of Figure 1.4 would change, and in what direction, if supply chain management systems were improved?
- 1.15 Discuss the potential interactions between the production planning and scheduling lever and the inventory management lever.
- 1.16 What new topics do you think will be included on the list of management levers in 10 years? What levers will be eliminated?
- 1.17 If it is true that U.S. firms focused on cost, then quality, and then on delivery and flexibility (i.e., time), what will they focus on next?
- 1.18 An important financial performance measure is the return on investment:

$$\text{ROI} = \frac{\text{Profit}}{\text{Total assets}}$$

Assume that inventories are 34% of current assets.

- a. Consider the typical company and assume that it has a ratio

$$\frac{\text{Current asset}}{\text{Total assets}} = 0.40$$

What impact will a 25% decrease in inventory have on the ROI? What percent decrease in inventories is needed to increase the ROI by 10%?

- b. Develop an algebraic expression for the impact on ROI of an  $x\%$  decrease in inventories when

$$\frac{\text{Current asset}}{\text{Total assets}} = f$$

- 1.19 Analyze the financial statements for the last 5 years of two companies from different industries. (E.g., a grocery chain and an electronic goods manufacturer.) Calculate their turnover ratios. Try to explain why their turnover ratios are different. Are there any trends apparent that can be explained?
- 1.20 Why is the turnover ratio widely used? What are its benefits and what are its drawbacks?
- 1.21 Blood is collected for medical purposes at various sites (such as mobile units), tested, separated into components, and shipped to a hospital blood bank. Each bank holds the components in inventory and issues them as needed to satisfy transfusion requests. There are eight major types of blood and each type has many components (such as red cells, white cells, and plasma). Each component has a different medical purpose and a different lifetime. For example, the lifetime for white cells is only 6 hours, but the lifetime for red cells is now 42 days. What type of management issues are involved in this particular type of inventory management situation? (Discuss from strategic, tactical, and operational perspectives.) Does perishability require different records than for other situations? How?

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## MANAGEMENT

In this section of the book, we turn to the details of decision making in a production environment

where one has to face two additional complexities. First, there are capacity constraints at individ

ual work centers, and this capacity must be shared among individual SKUs (as was discussed in

portions of Chapter 10). Second, items are closely related in that raw materials are converted into

final products through some production process. Thus, coordination of decisions about individual

SKUs is mandatory. In Chapter 13, we provide a broad overview of production planning and scheduling. A deci

sion framework is presented that encompasses strategic (long-range), tactical (medium-range), and

operational (short-range) concerns. Chapter 14 addresses medium-range aggregate production planning. This encompasses the

selection of work force levels and aggregate production rates, typically month by month out to

a horizon of approximately 1 year. Particular emphasis is given to some of the nonmathematical

aspects of aggregate planning. In Chapter 15, we turn to assembly situations where the overriding concern is with proper coor

dination of materials and effective use of labor. Specifically, the details of the Material Requirements

Planning system are presented. Chapter 16 examines the popular JIT system and a related system called OPT, and introduces

short-range sequencing and scheduling decisions. The following framework that was introduced in Chapter 2 is repeated here with relevant

chapter numbers included. Product mix Process pattern Job shop (very jumbled flow) Sequencing rules; factory physics (Chapter 16) OPT (Chapter 16) MRP and JIT (Chapters 13, 15, and 16) JIT (Chapter 16) Periodic review/ cyclic scheduling (Chapter 10) Batch flow (less jumbled) Worker-paced line flow Machine-paced line flow Continuous, automated, rigid flow Few of each; custom Low volume; many products Very high volume; commodity High volume; several major products

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## Appendix III: Approximations and Excel Functions

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